Aryan Shrivastava

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3rd-year Integrated M.Sc student, Department of Physics,
National Institute of Science Education and Research (NISER)
Bhubaneswar, Odisha, India

Objective

To work in scientific environments where I can apply my skills in physics and machine learning. Passionate about mathematics and enjoy solving complex problems. Aim to contribute to innovative research and gain valuable hands-on experience.

Achievements

- Recipient of Innovation in Science Pursuit for Inspired Research **(INSPIRE)** scholarship.
- Silver medal in the National Competition for Computational Physics (organized by the Indian Association of Physics Teachers).
- Outstanding Performance award in 3rd Semester AY 2024-25 with a SGPA of 9.43.
- Outstanding Performance award in 4th semester AY 2024-25 with a SGPA of 9.09.

Projects

Muon Energy Reconstruction in INO-ICAL Detector using Machine Learning See details

This project involves a benchmark for 3 ML models: Artificial Neural Network, Random Forest and XGBoost model for the regression task of energy reconstruction of Muons from a GEANT4 simulated dataset based on INO-ICAL Detector.

Physics Enhanced Neural Networks for Chaotic Hamiltonian Dynamics See details

This project implements a Physics-Enhanced Neural Network architecture that uses the Hamiltonian of a system in the model. It achieves a better long-term prediction stability in phase-space trajectories compared to standard artificial neural networks.

Experience

- Attended the workshop "Quantum Quest: Principles, Security and Applications" organised by AACST (Association of All Computer Science Teachers)
- Summer Internship under the guidance of Dr Moon Moon Devi, at Tezpur University sponsored by the IASc - INSA - NASI Summer Fellowship Programme on **Particle Physics: Simulations and Machine Learning**
- An NPTEL course on Deep Learning by IIT Kharagpur
- A Machine learning specialization by Andrew Ng on Coursera
  - Supervised learning: regression and classification
  - Advanced learning algorithms

Education

2023 - present	Integrated M.Sc at **National Institute of Science Education & Research (NISER)** (Currently in 6th semester)	(GPA: **8.98**/10)
2023	12th CBSE Board at **Chavara Vidyapeeth, Narsinghpur**	(94.8%)
2021	10th CBSE Board at **Chavara Vidyapeeth, Narsinghpur**	(93%)

Technical Skills

Programming Languages	Python (Proficient in TensorFlow, Pandas, Scikit-learn, Matplotlib, Seaborn)
Machine Learning & Data Analysis	Experienced in developing Machine Learning models and skilled in data preprocessing, visualization, and performance evaluation.
Software and Tools	LaTeX, Git, MS Office, Google Workspace, Jupyter Notebook
Relevant Math Courses	Linear Algebra, Probability Theory, Metric Spaces & Differential Equations

Research Interests

- Scientific Machine Learning
- Quantum Mechanics
- High Energy Physics
- Non-linear Dynamics
- Computational Physics
- Mathematical Physics

Extra-Curricular Activities

Sports	Cricket & Chess
Club	Member of Physics club (Vikiran), NISER

Certificates & Grade Cards

All the certificates and academic transcripts can be accessed via the link below:

Certificates & Grade Cards